

Máster de Formación Permanente

Máster en Energías Renovables y Medio Ambiente

20
EDICIÓN
2025-2026



UNIVERSIDAD
POLITÉCNICA
DE MADRID

escuela técnica superior de
ingeniería
diseño industrial



Módulo 1
Mercado Energético

Renewable Power Purchase
Agreement (PPA)

Marta Arias Álvarez



Marta Arias Álvarez



<https://www.linkedin.com/in/marta-arias-a824639/>



marta.arias@edp.com



Contents

- | | |
|----|--------------------------------|
| 01 | PPAs introduction |
| > | What is a PPA? |
| > | What is a PPA for? |
| > | Applicable regulation |
| > | Vocabulary tips |
| 02 | Renewable PPAs |
| 03 | PPA process |
| 04 | Digital platforms for PPAs |
| 05 | PPAs trends and market outlook |



A Power Purchase Agreement (PPA) is a contract between an energy buyer (*off-taker*) and seller (*generator, independent power producer, investor*) to purchase energy generated by a renewable asset at pre-agreed prices for pre-agreed periods – *whether the asset already exists or is planned*.

PPAs can be signed for a short-term period, which is usually defined as 2 to 5 years, or a long-term period between 5 and 20 years.

KEY TAKEAWAYS

1.

Power Purchase Agreement (PPA) is a **contract** which has two parties involved: buyer and seller **to purchase electricity**

2.

The electricity can be supplied by **existing** renewable energy assets **or new build projects**. Newly built projects often have more stringent terms

3.

New project deals are typically **long-term (10+ years)** PPAs, but short-term PPAs or PPAs for existing assets can have a tenor of **less than a year**.

Contents

01 PPAs introduction

- > What is a PPA?
- > What is a PPA for?
- > Applicable regulation
- > Vocabulary tips

02 Renewable PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook



01 PPAs introduction – What is a PPA for?



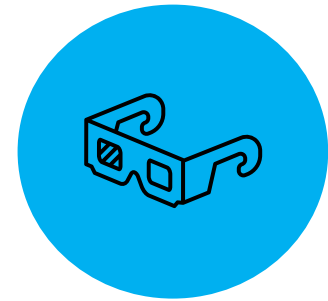
Project Financing

- Renewables often need a **third-party funding source**, such as a bank. It is, however, unlikely that a third party will lend without a security.
- In the absence of a government subsidy, a wind or solar PPA **provides that assurance**.



Hedging

- PPAs allow a portion of a project's **future energy generation to be sold** to an energy buyer over a pre-defined and fixed period – typically between 2 and 30 years. The contract is usually agreed upon and signed before the project begins.



Long-term Price Predictability

- Electricity prices can fluctuate greatly and frequently. The main characteristic of a power purchase agreement is the agreement **to sell X amount of MWh** from a renewables project to a buyer of energy **at a fixed price**.

Contents

01 PPAs introduction

- > What is a PPA?
- > What is a PPA for?
- > Applicable regulation
- > Vocabulary tips

02 Renewable PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook





COMMISSION RECOMMENDATION

of 18.5.2022

on speeding up permit-granting procedures for renewable energy projects and facilitating Power Purchase Agreements

WHAT IS THE OBJECTIVE OF THIS **RECOMMENDATION**?

Renewable energies are central to the energy transition needed to achieve the goals of the European Green Deal, make energy affordable, and reduce the European Union's (EU) dependence on fossil fuels and energy imports.

The Recommendation aims to:

- speed up permitting procedures for renewable energy projects, and
- facilitate power purchase agreements (direct contracts between companies and electricity suppliers).

<https://eur-lex.europa.eu/ES/legal-content/summary/fast-tracking-renewable-energy-projects.html>

01 PPAs introduction – Applicable regulation

Regulation (EU) 2024/1747 (la reforma del diseño del mercado eléctrico de la UE

<https://eur-lex.europa.eu/eli/reg/2024/1747/oj/eng>



Aspecto

Reforma introducida

Impacto esperado

Promoción de los PPAs

Los Estados miembros deben promover el uso de PPAs facilitando instrumentos que reduzcan el riesgo de impago de los compradores (“offtakers”), por ejemplo, mediante esquemas de garantía accesibles para clientes que enfrentan barreras para acceder al mercado de PPA.

Busca **eliminar uno de los obstáculos financieros más comunes** (riesgo de contraparte) para que empresas, consumidores industriales o medianos puedan contratar PPAs con generadores renovables.

Plantillas voluntarias de PPA

La Agencia para la Cooperación de Reguladores de Energía (**ACER**) evaluará la necesidad de plantillas voluntarias estandarizadas de PPA adaptadas a diferentes tipos de contrapartes, con fecha límite para el 17 octubre 2024.

Facilita la negociación y reduce la carga de transacción y coste legal, especialmente para compradores menos sofisticados.

Instrumentos de apoyo financiero para generadores

A partir del 17 de julio de 2027, los incentivos de apoyo público para nueva generación (eólica, solar, hidrológica sin embalse, geotérmica y nuclear) ya no podrán ser, en general, pagos unilaterales, sino que deberán adoptar la forma de contratos por diferencia bidireccionales (two-way CfD) o esquemas equivalentes que garanticen reparto de riesgos entre el Estado & el generador.

Esto cambia el **equilibrio de riesgos** en nuevas inversiones: los generadores estarán más expuestos a las fluctuaciones del mercado, pero recibirán compensación cuando los precios estén bajos, y devolverán cuando sean muy altos.

01 PPAs introduction – Applicable regulation

Regulation (EU) 2024/1747 (la reforma del diseño del mercado eléctrico de la UE



Aspecto

Reforma introducida

Impacto esperado

Objetivos indicativos de flexibilidad no fósil / almacenamiento

Se introduce la exigencia de que los Estados miembros evalúen cada dos años sus necesidades de flexibilidad no fósil (respuesta de demanda, almacenamiento, hidro con embalse) y definan objetivos indicativos, con posibilidad de esquemas de apoyo a la capacidad disponible de flexibilidad

En mercados con alta penetración de renovables, la gestión del balance entre generación y demanda será crítica, y los PPAs podrían incorporar cláusulas relacionadas con **flexibilidad, curtailment, almacenamiento**, etc.

Mercados forward y “regional virtual hubs”

Se incorpora en la regulación la posibilidad de crear “regional virtual hubs” (RVH) para mercados de electricidad forward (es decir, hubs virtuales de referencia regional).

Un hub regional de precios forward puede facilitar la liquidación de contratos de cobertura (incluyendo PPAs) en un punto de referencia común, reduciendo la fragmentación de precios entre zonas eléctricas.

Evaluación y supervisión del mercado PPA

Se prevé una evaluación anual de los mercados de PPA a nivel de la Unión y de los Estados miembros.

Esto permitirá **monitorear** avances, identificar obstáculos persistentes y ajustar políticas nacionales/regulatorias en consecuencia.

01 PPAs introduction – Applicable regulation



Official Journal
of the European Union

EN
L series

2024/1747

26.6.2024

REGULATION (EU) 2024/1747 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 13 June 2024

amending Regulations (EU) 2019/942 and (EU) 2019/943 as regards improving the Union's electricity
market design

'CHAPTER IIIa

(Text with EEA relevance)

SPECIFIC INVESTMENT INCENTIVES TO ACHIEVE THE UNION'S DECARBONISATION OBJECTIVES

Article 19a

Power purchase agreements

- It is expressly provided that Member States must ensure the existence of guarantee schemes at market prices aimed at reducing the financial risks associated to offtaker payment default under PPAs, specifying that these instruments may include state-backed guarantee schemes at market prices, private guarantees, or facilities pooling demand for PPAs.

Article 19b

Voluntary templates for PPAs and monitoring of PPAs

- The need to develop and issue voluntary templates for PPAs, adapted to the needs of the different categories of counterparties.
- <https://www.acer.europa.eu/electricity/market-monitoring/ppas>

Article 19d

Direct price support schemes in the form of two-way contracts for difference for investment

- It is envisaged that direct price support schemes for investments in new power-generating facilities for the generation of electricity from wind, solar, geothermal, hydroelectric without reservoir, and nuclear energy sources will take the form of two-way contracts for difference for investment or equivalent regimes that produce the same effects



01 PPAs introduction – Applicable regulation

Tecnología	Potencia Adjudicada (MW)
Fotovoltaica	2.036
Eólica	998



WHO IS THE BUYER? SPANISH ELECTRICITY MARKET (OMIE)

1

The auction is carried out according to a **fixed price** per unit of energy (€/MWh),

2

Seller will always **sell on the market**, in a period between the start and end of the "maximum delivery period"

3

The buyer (OMIE), the market operator, is responsible for calculating the process energy.

...AND WHO IS THE SELLER AND WHAT TECHNOLOGY?

Fotovoltaica		
Empresa	Potencia (MW)	Horquilla de precios (€/MWh)
X-ELIO	315	19,8-27,8
IBERDROLA	243	22,87-25,87
NATURGY	197	23,45-25,68
SOLARIA	180	27,91-28,05
ELAWAN	175	23,98-27,98
IB VOGT	150	23,94
GRUPO IGNIS.	125	14,89-23,49
ACCIONA	106,6	19,44-27,29
EDP RENOVABLES	98	18,99-27,01
HANWHA ENERGY	86	26,77-27,89
ENGIE ESPAÑA	85	23,49-25,89
AKUO RENOVABLES	81	23,5-27,5
ENDESA	50	28,9
DOMINION ENERGY	45	24,93
FALCK RENEWABLES	40	24,79
RIOS RENOVABLES	30	26,99-28,74
NRG	30	25-27
Q-ENERGY	20	28,7
CANADIAN SOLAR	14	25,2
PROSOLIA	10	27
ALTER ENERSUN	8	26,9
LIGHTSOURCE BP	5	23,97
CUBE ENERGY	1	25

Eólica		
Empresas	Potencia (MW)	Horquilla de precios (€/MWh)
CAPITAL ENERGY	620	23,86 - 28,6
GREENALIA	134,3	28,5 - 29
ELAWAN	105	22,88 - 27,98
EDP RENOVABLES	45	24,99
ENERFÍN	40	20
NATURGY	37,95	28,63
EURUS	14	26,5

Auctions, as their name suggests, are processes that award winning companies the opportunity to execute a project, in this case, a facility with renewable technology.

Contents

01 PPAs introduction

- > What is a PPA?
- > What is a PPA for?
- > Applicable regulation
- > Vocabulary tips

02 Renewable PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook



01 PPAs introduction – Vocabulary tips



Offtaker

- is another name for **energy buyer**. As used in project financing, this is the party **who buys** the product being produced by the project or uses the services being sold by the project (for example, electricity, mined copper, or a pipeline)

Open markets

- When talking about 'open markets', we mean **the absence of government involvement**, i.e., the absence of subsidies.

Third-party

- Third-party **lenders** can be banks, lenders, credit providers or finance providers.

GoO / GO

- Renewable attributes are energy credits, renewable energy certificates, Guarantees of Origin (GoOs). Guarantees of Origin are the only defined instruments evidencing the origin of electricity generated from renewable energy sources.

PaP

- Pay as Produce is an agreement where the **payment** to the energy producer is based on **the actual energy generated and delivered**.

C&I

- **Corporate and Industrial** buyers

CfD

- **Contract for Difference**: A contract where two parties agree to settle the difference between a fixed price (the strike price) and the prevailing market price of an asset

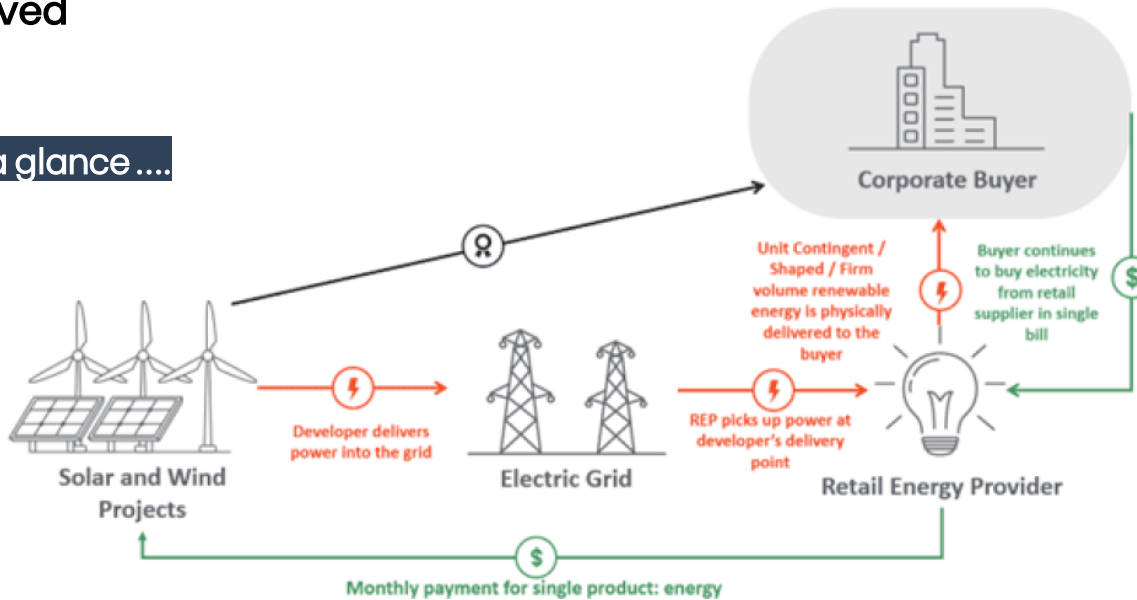
Contents

- 01 PPAs introduction
- 02 Renewable PPAs
 - > Parties involved
 - > Objectives of the parties
 - > Types of PPAs
 - > Contractual information
 - > Risk in PPAs
- 03 PPA process
- 04 Digital platforms for PPAs
- 05 PPAs trends and market outlook



02 Renewable PPAs – Parties involved

Renewable PPA at a glance



... and two key parties are involved

<https://www.trioadvisory.com/resources/retail-renewable-ppas-101-heres-what-you-need-know>

Two key parties are involved in a typical PPA: Energy

1. **Seller – Generator or Producer**
2. **Buyer or Offtaker**
 - Utility (utility PPA),
 - Corporate consumer (corporate PPA)

Seller | Generator | Producer

Usually a company that **develops, builds, and operates a renewable energy farm** (solar, wind, hydro, etc.).

Financiers and Investors (*often indirectly*) Banks, investment funds, or financial institutions that provide the capital needed to build the project. They are not always signatories to the PPA, but the contract is a key condition for securing financing.

Buyer – Offtaker

- This can be an **electric utility – retail energy provider** (utility PPA), which purchases the energy to resell it to end customers,
- or a **corporate consumer – corporate buyer – C&Is** (corporate PPA), such as Google, Amazon, or a large industry that wants to source renewable electricity. Their interest is to fix the price of energy over time and meet sustainability objectives (e.g., decarbonization or ESG).

Contents

- 01 PPAs introduction
- 02 Renewable PPAs
 - > Parties involved
 - > Objectives of the parties
 - > Types of PPAs
 - > Contractual information
 - > Risk in PPAs
- 03 PPA process
- 04 Digital platforms for PPAs
- 05 PPAs trends and market outlook



02 Renewable PPAs – Objectives of the parties



Buyer



Principal

- Ensure a competitive and stable supply of renewable energy
- .Protect yourself against wholesale market volatility.

Secondary

- Meet sustainability targets and ESG/Net Zero commitments.
- Differentiate your commercial offering (utility) or your end product/service (corporation) as “green.”
- Flexibly manage your portfolio (part covered by PPAs, part exposed to the market).



Seller



Principal

- Ensure stable long-term income (cash flow predictability).
- Guarantee the project's bankability (a necessary condition for obtaining financing).

Secondary

- Maximize the price obtained compared to other alternatives (merchant, other PPAs).
- ransfer risks (market, regulatory, credit) to the buyer to the greatest extent possible.
- Enhance reputation as a green energy supplier.



Investors



Principal

- Have a solid, long-term contract that reduces the risk of non-payment.
- Validate the solvency of the counterparty (utility or corporation).

Secondary

- Minimize legal, technical, and regulatory risks.
- Ensure competitive returns compared to other investments.



Long term position management is just a ‘grocery shop’, where the objective is to buy at low price and sell at high price

Contents

01 PPAs introduction

02 Renewable PPAs

- > Parties involved
- > Objectives of the parties
- > Types of PPAs
- > Contractual information
- > Risk in PPAs

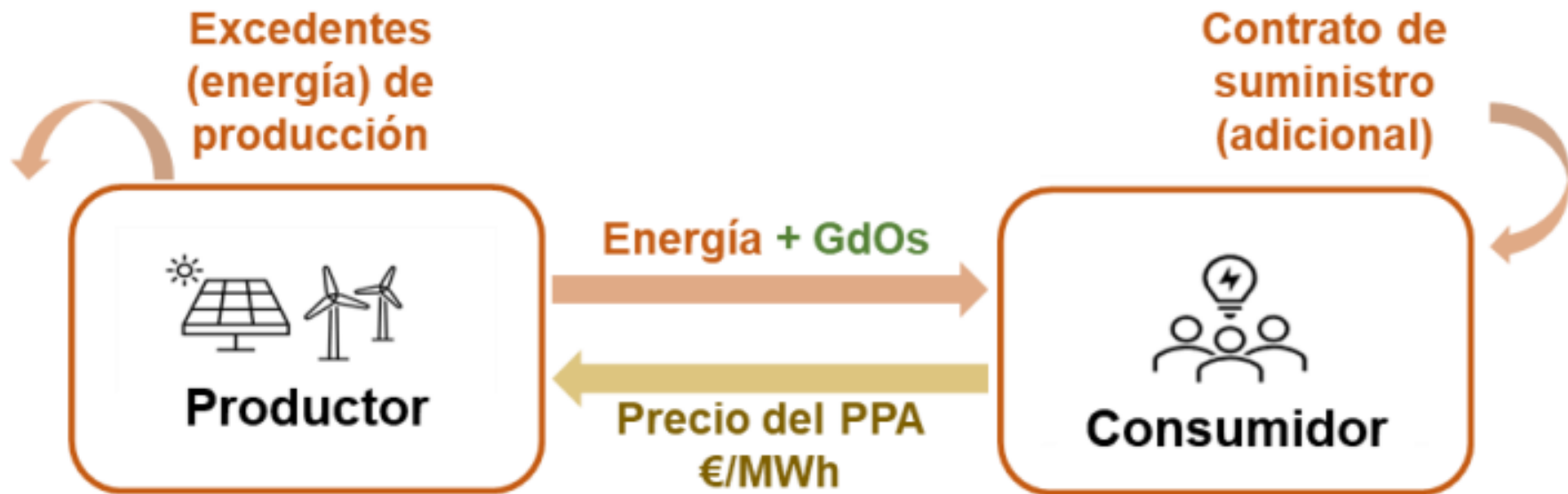
03 PPA process

04 Digital platforms for PPAs

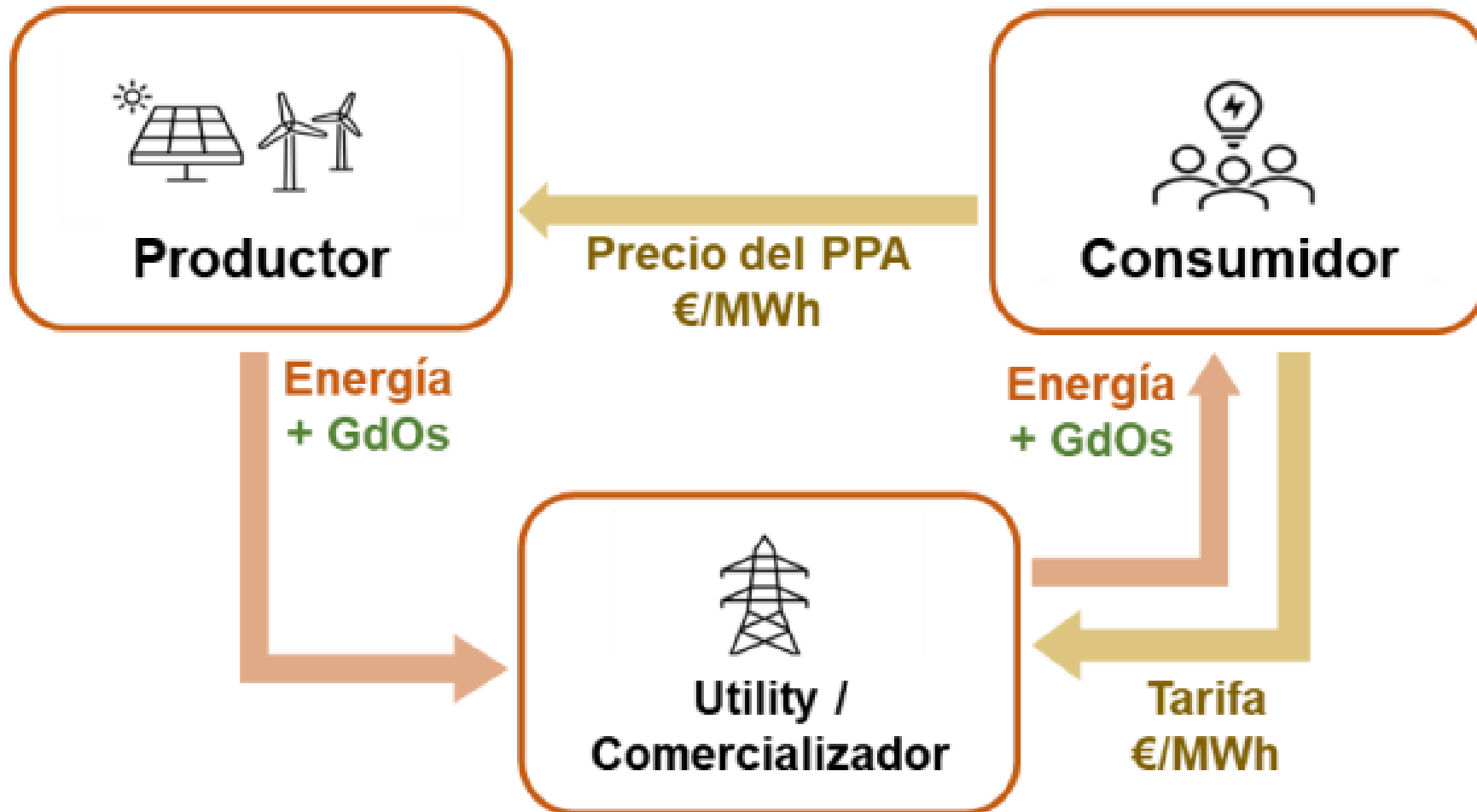
05 PPAs trends and market outlook



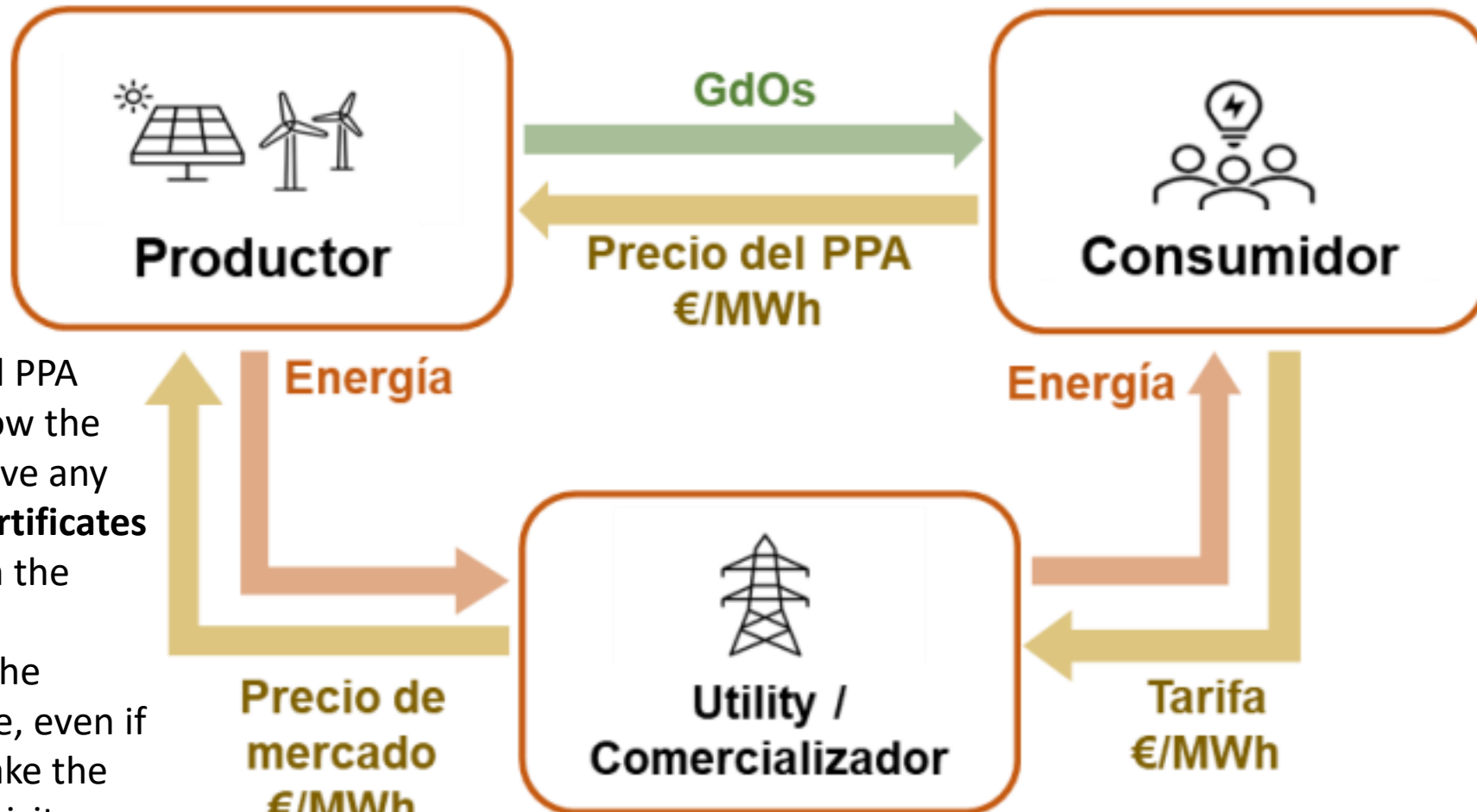
PHYSICAL PPA ON SITE



PHYSICAL PPA OFF SITE (SLEEVED)



VIRTUAL PPA



What a virtual PPA does do is allow the buyer to receive any **renewable certificates** resulting from the production of electricity at the renewable site, even if they do not take the physical electricity itself.

Contents

01 PPAs introduction

02 Renewable PPAs

- > Parties involved
- > Objectives of the parties
- > Types of PPAs
- > Contractual information
- > Risk in PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook



The aim of the section, therefore, is to provide you with an overview of the main **contractual areas** of a PPA, while also showing you how these areas are represented in an official PPA.

CONTRACTUAL CONCEPTS

1.

The **commercial terms** are usually the first section of the PPA and will contain: type of PPA, price, quantity of electricity, time period for delivery and the Renewable Certificates

2.

The **Asset Information section** of the Renewable PPA will contain such details as: COD, forecasting, outages and maintenance, etc.

3.

Key **Legal Considerations and financial Information** as reasons for termination, insurance, credit support, etc.

02 Renewable PPAs – Contractual information – Commercial Terms

This section looks at the information contained within the **commercial terms** section of the PPA. The commercial terms are usually the first section of the PPA. The section outlines:

TYPE OF PPA	PPA PRICE	TIME PERIOD	QUANTITY OF ELECTRICITY	RENEWAL CERTIFICATES
Physical PPA (sleeved or not) Virtual PPA	Fixed and variable (always clearly referenced)	Start, date and duration of the contract	Quantity of expected electricity is explicitly stated	These renewable certificates can be traded in the market
Just a reminder: • Physical PPA there will be a physical exchange of electricity such that the buyer will accept delivery of electricity produced by the renewable asset owned by the seller • Virtual PPA where no physical exchange of electricity takes place	Some examples: • Fix price price to be paid for each unit of electricity produced • Index price marked against some form of market index price • Hybrid price Fixed + Index • CfD (Contract for Difference) buyer and seller pay each other the difference at market price	Define the duration of the contract: • Generally, pays will range from 1 to 15 years. • Within this time frame you may also have seasonal PPAs whereby a solar farm, for example, supplies power over a 15 year period but does so only during summer months	Aligned business needs with application usage: • As generated estimated yearly output figure from the renewable asset (seller) • As consumed the forecast consumption from the buyer • Baseload single number, is agreed in the contract and this is the volume supplied for the duration of the PPA	The fundamental structure of renewable certificates tend to be the same, with one MWh of production of electricity from a renewable asset generating one renewable certificate which is supplied from the government or some other energy related authority.

02 Renewable PPAs – Contractual information – Asset information



This section looks at the information contained within the **asset information** section of the PPA. The Asset Information section of the Renewable PPA will contain such details as:

BASIC ASSET DETAILS	COD	FORECASTING	OUTAGES AND MAINTENANCE	ACCESS TO INFORMATION
Basic information of the renewable asset	Commercial Operations Date (asset start date)	Predict what the electricity output from the asset	Schedule of any planned outages at the asset	Metering and real-time generation of the asset
The basic asset details: • Asset name • Asset generation type: Wind, solar, etc. • Capacity listed in MW. what is the maximum number of megawatts it can produce? • Site address, • A network operator who owns and operates the electricity network that the asset is connecting into.	Key information: • This date is extremely important to the buyer. The buyer will have based its financial models on this date • The seller will generally have to pay the buyer a daily fee for each day that the asset does not achieve COD. These fees are commonly termed liquidated damages.	• This is usually provided on a yearly basis at the start of the calendar year, advising the buyer what they forecast the asset to produce each day of that year. • The forecasts can be granular with the forecast provided for an entire year that predict what each half hour of that year will look like.	Seller must provide this information : • It is standard practice for renewable assets to have at least one planned outage per year. • There is also an expectation that any changes made to the forecast or planned outage schedule are communicated to the buyer in a timely manner	This will be fed directly from the asset via an API or some other form of connection directly to the buyer who will use this information to monitor what output the asset is producing.

This section looks at the information contained within the **Key legal considerations & Financial information** section of the PPA.

CONDITIONS PRECEDENT

Tasks that the seller and the buyer **must complete ahead of** the live operation of the asset

For the **seller**.

- To obtain all relevant consents and permits relating to the construction of the site.
- To have all the necessary land and access rights to the facility.
- To have signed a connection agreement with the relevant network operator.

For the **buyer**.

- providing suitable credit support.
- And having obtained the relevant regulatory approvals required

MATERIAL REASONS FOR TERMINATION

Reasons for termination of the PPA contract

Common material reasons for terminating the contract are:

- **Failure to deliver or accept the electricity** of renewable certificates.
- **Missed payments**: will generally be followed by a set of timelines,
- **Representation or warranty**. covers situations where either party may have lied or misled the other party relating to either their financial standing or their capabilities as a company.

FINANCIAL INFORMATION

Settlement and PPA contract guarantees

- **Invoicing and Billing** – where either the Buyer or the Seller has the responsibility of sending out invoices, to which the other party must pay within a defined timescale
- **Credit Support** – where each party must ensure they have sufficient credit to cover any potential payments during the course of the agreement
- **Insurance** – where both parties must prove they are suitably insured to cover any accidental damage to the renewable asset

Contents

01 PPAs introduction

02 Renewable PPAs

- > Parties involved
- > Objectives of the parties
- > Types of PPAs
- > Contractual information
- > Risk in PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook



The aim of this section is to provide you with an understanding of the **key risks** involved in a renewable

KEY RISKS IN A PPA

1.

The **project development phase** of a renewable asset and what the implications of these risks are for any potential power purchase agreement.

2.

The **power price** is impacted by: the **demand**, the **cost** or availability **of the fuel** used to produce power, **outages** at power plants, etc.

3.

The **traders** who worked for both the **sellers** of PPAs and for or on behalf of the **buyers** of PPAs can buy power on a power trading exchange.

4.

The level of risk for the Buyer and the Seller varies depending on which **type of volume delivery** is used

PROJECT DEVELOPMENT PHASE



Building a renewable asset takes considerable time and planning

Asset developers must secure:

- **land to build** the asset on the most secure planning
- **permission from local or national authorities**, giving them the right to build the asset, and they
- must secure a **connection agreement with the electricity grid operator**, allowing them to connect the site to the electricity network on a specific date.



- **Negotiate/Close a PPA** aligned with the COD of the renewable asset under construction

Risk of this phase

Technical and Commercial time alignment	• There is significant risk to the asset developer that their commercial timelines, i.e. looking to secure a PPA under development and pre-construction timelines, i.e. securing land and materials don't match up.
Commodity Prices	• Commodity prices of such materials as steel and aluminium rose significantly from 2021 into 2022, with material costs rising as much as 60 and 70%.
Network connection	• The asset developer is still awaiting confirmation on the connection agreement from the electricity network operator and the subsequent cost to them of connecting to the network, this can be millions of dollars depending on the location

Note here that in all three examples, **the risks don't disappear**. They just tend to be moved from the buyer to the seller or vice versa, and at times split between the two parties

Risk involved

POWER PRICE



Different variables that impact the power price.

Different variables that impact the power price:

- Namely **demand** for power
- the **cost or availability** of the main fuel input into power production
- the **availability of power generation** and the impact of international markets
- power prices in a particular country may be impacted by **power prices in neighboring countries**.

Demand for power

- The standard demand curve for electricity tends to have two peaks.
 - One in the morning as people wake up, use the kettle and toaster and get ready to start work.
 - The second peak follows later, when people return home from work and make dinner and switch on lights and the television
- **Power prices tend to be more expensive** during times when the **demand** for power is **higher**.

Availability of the fuel used

- **Gas prices** can significantly impact the price of power. In many countries where there are a large number of power plants that use gas as their source of fuel.
- In a market where **renewable generation** makes up a greater percentage of the power production mix

Availability of power generation

- Countries who operate a large number of nuclear plants, for example, may see high power prices during periods where their nuclear plants are on outage.
- Without their generation capacity, a country may be forced to turn on other, more expensive types of power generation leading to higher power prices.

International markets

- This is particularly true if a country happens to be a **net importer of power** from other countries.

Risk involved

TRADING POWER



A power trading exchange
is a place where traders
can trade a number of
different types of trade

The traders who worked for both

- the **sellers** of PPAs
- and the traders who work for or on behalf of the **buyers** of PPAs can buy power on a power trading exchange.

Liquidity

- Liquidity in this sense can be defined as having lots of buyers and sellers.
- Trading these particular products. the most liquid power markets tend to be those closer to today's date. This indicates that all things equal, it is easier to buy power for the next day or even a few months ahead than it is to buy power years in advance.

Availability produce power

- Market players who enter into PPAs, however, do so because they are more certain about their ability to supply power, i.e. we are going to build an asset that will produce X amount of power over a long period of time.

Availability buy power

- If we have a long term plan, be it manufacturing firms, building production plants or tech firms building data centers, and we need X amount of power.

PRICE AND VOLUME RISK



All **three types** experience price and volume risks.

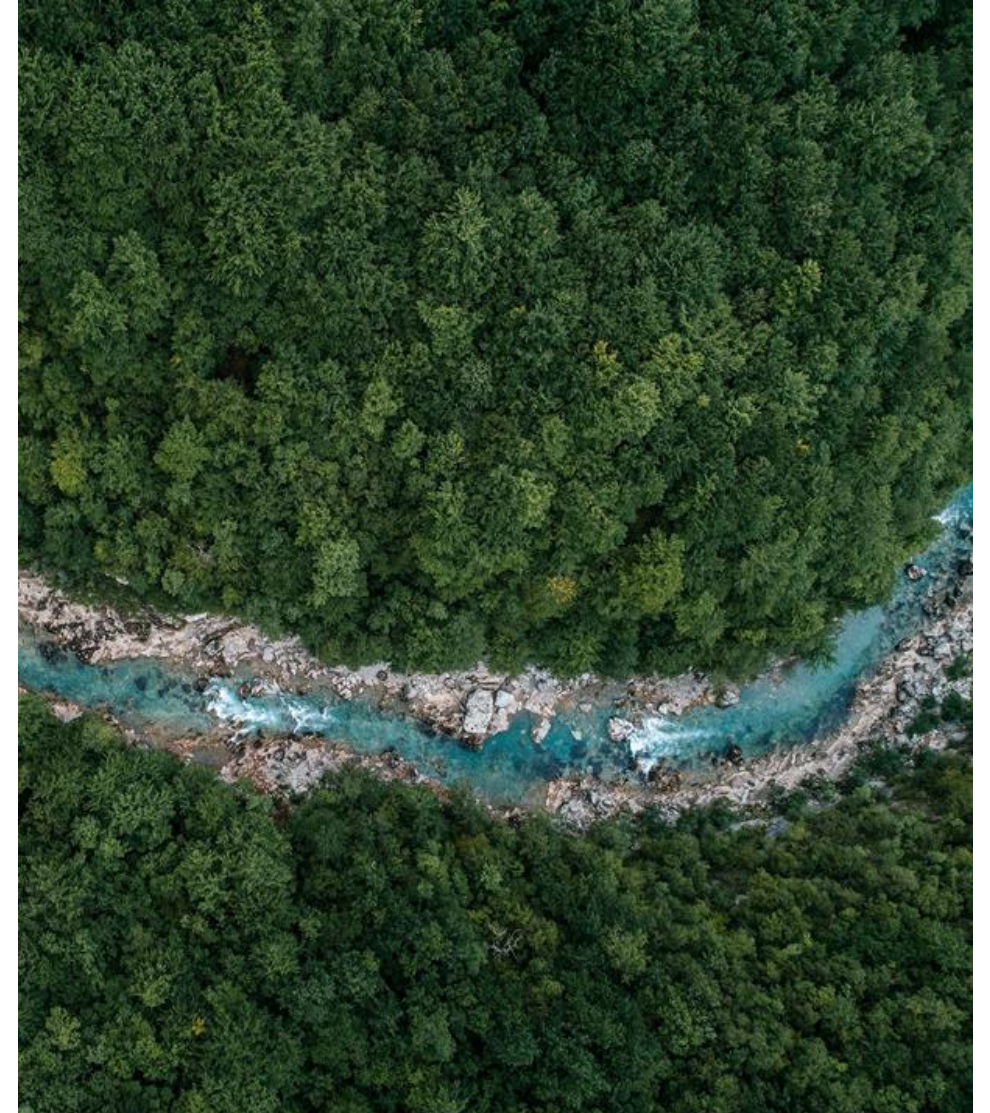
- For the **AS GENERATED** PPA, the bulk of the volume risk lies with the **Buyer** who will have to procure power to match their usage.
- For **AS CONSUMED** PPA's, the volume and price risk lies with the **Seller** who will potentially be buying power at times of high prices and selling power at times of low prices.
- The **BASELOAD** PPA encompasses both of these risks, but they are easier to manage, with the seller able to manage these risks

Risk involved

As Generated	• The bulk of the volume risk lies with the buyer who will have to procure power to match their usage, where they will most likely have to buy additional power at expensive times when there is a lower level of renewable energy production.
As Consumed	• Conversely for as consumed PPA, the volume and price risk lies with the seller who will potentially be buying power at times of high prices and selling power at times of low prices.
Baseload	• The base load PPA encompasses both of these risks, but they are easier to manage with the seller able to manage these risks at least in some part years in advance by trading against a fixed volume, i.e. the base load volume.

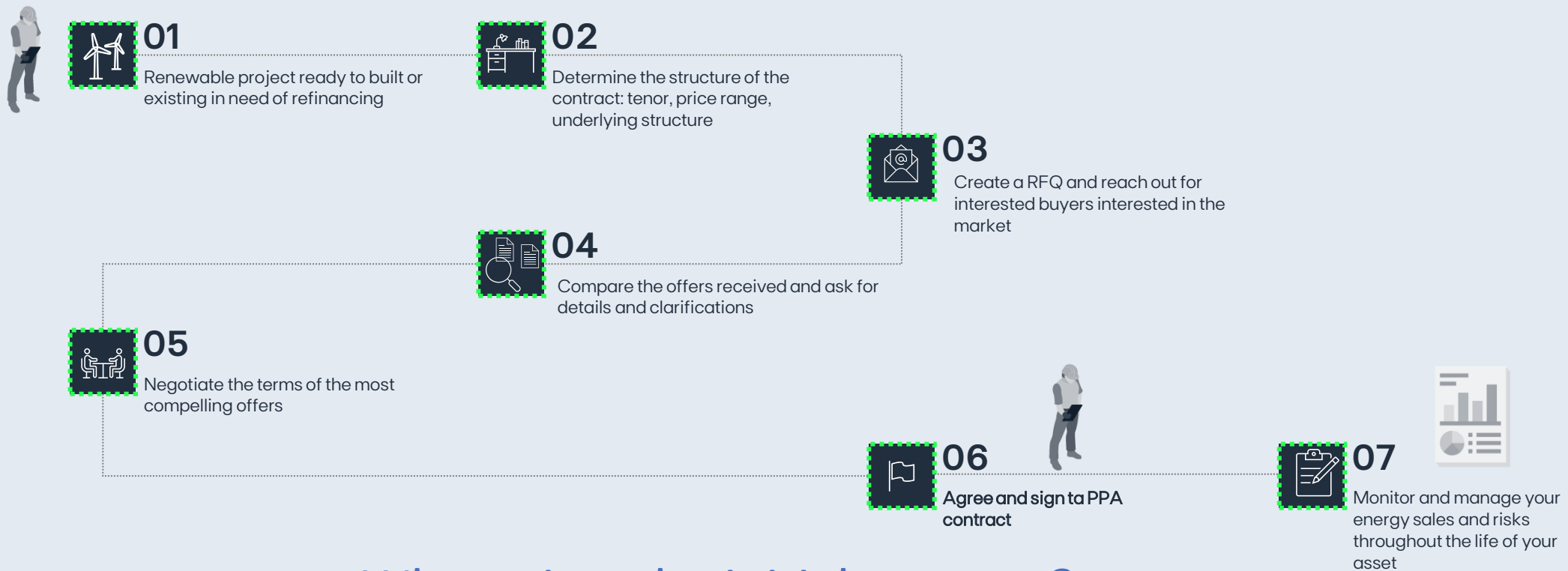
Contents

- 01 PPAs introduction
- 02 Renewable PPAs
 - > Parties involved
 - > Objectives of the parties
 - > Types of PPAs
 - > Contractual information
 - > Risk in PPAs
- 03 PPA process
- 04 Digital platforms for PPAs
- 05 PPAs trends and market outlook



03 PPA Process

The following outlines a simplified version of the PPA process:



Who writes the initial contract?

Contents

01 PPAs introduction

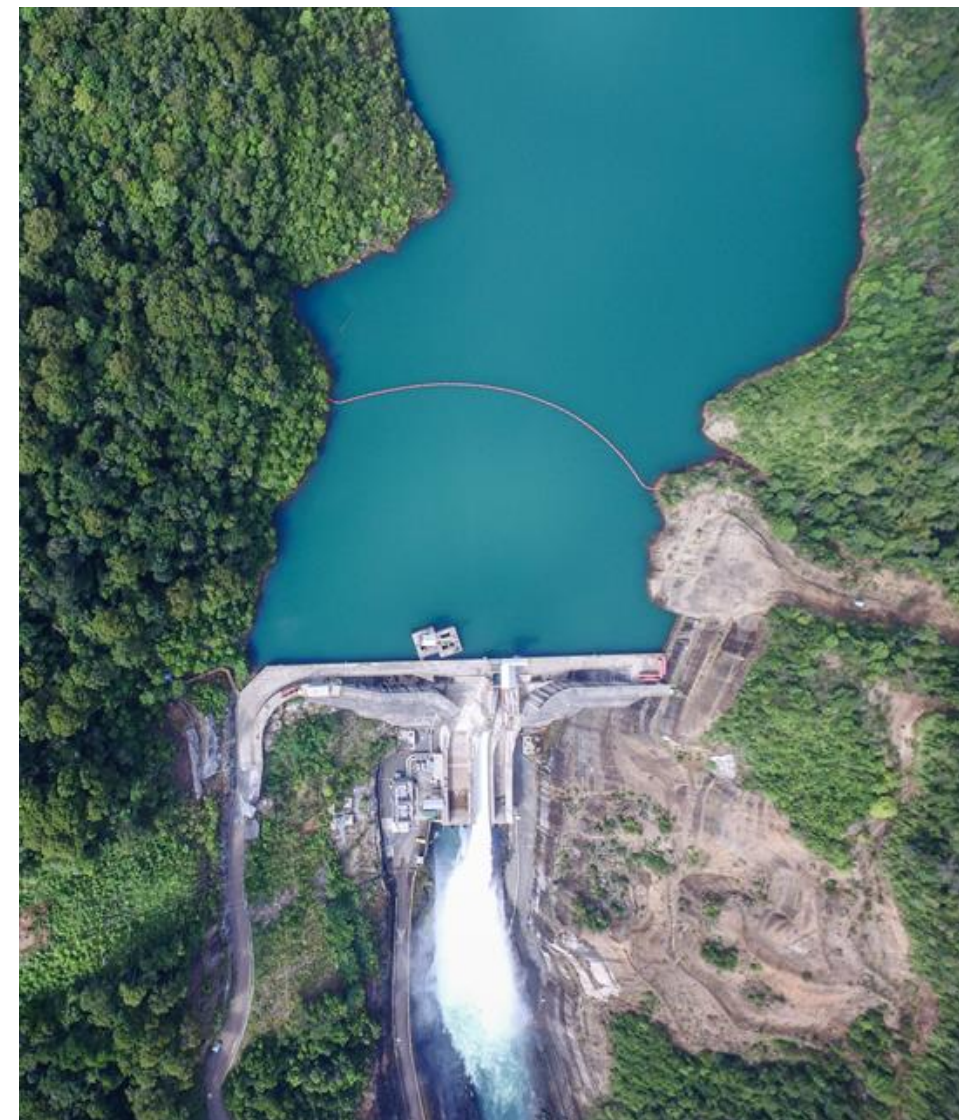
02 Renewable PPAs

- > Parties involved
- > Objectives of the parties
- > Types of PPAs
- > Contractual information
- > Risk in PPAs

03 PPA process

04 Digital platforms for PPAs

05 PPAs trends and market outlook



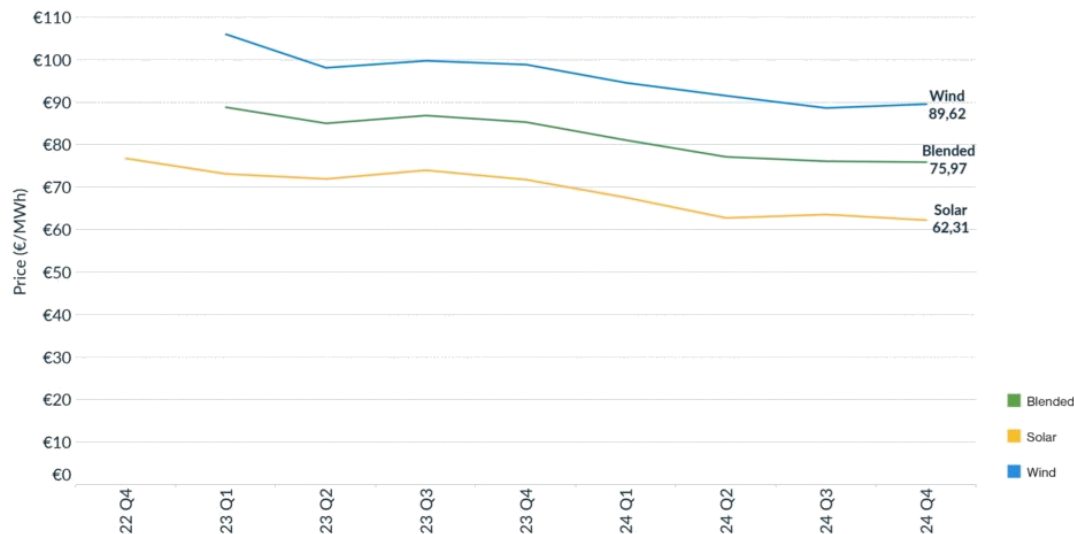
04 Digital platforms for PPAs

1. PPA Prices

Due to a general lack of transparency in the PPA market, PPA prices in Europe are not readily available. In fact, the process of getting them is cumbersome. You would normally need to contact potential buyers, outline your project, sign NDAs (*Non-Disclosure Agreements*) and maybe get a price range. <https://www.leveltenenergy.com/platform/market-transparency-reports>



Q4 2024 PPA Price Index for Europe



Market-Averaged Continental Index: Q4 2022 to Q4 2024

Precios de ofertas de PPA Solar P25 por país:

Índice	Precio Q4 2024	Cambio Euro Trimestral, 2024 Q3 a 2024 Q4	Cambio Porcentaje Trimestral, 2024 Q3 a 2024 Q4	Cambio Euro Interanual, 2023 Q4 a 2024 Q4	Cambio Porcentaje Interanual, 2023 Q4 a 2024 Q4
Bulgaria	€63,49	€0,99	1,6%	-	-
Dinamarca	€49,96	-	-	-	-
Finlandia	€46,63	-	-	-€8,38	-15,2%
Francia	€67,00	-	-	-€15,74	-19,0%
Alemania	€63,25	-€1,75	-2,7%	-€13,75	-17,9%
Italia	€65,00	-€3,50	-5,1%	-€9,36	-12,6%
Polonia	€78,00	-€5,00	-6,0%	-€13,22	-14,5%
Portugal	€38,50	-€1,50	-3,8%	-€7,75	-16,8%
Rumania	€63,00	-€0,28	-0,4%	-€6,97	-10,0%
España	€38,97	-€0,03	-0,1%	-€4,03	-9,4%
Suecia	€43,00	-	-	-€21,50	-33,3%
Reino Unido	€90,92	-€0,20	-0,2%	-€9,13	-9,1%

1. RE-Source Platform

The RE-Source Platform is a European alliance of stakeholders representing clean energy buyers and suppliers for corporate renewable energy sourcing.

This platform pools resources and coordinates activities to promote a better framework for corporate renewable energy sourcing at the EU and national levels.

Learn more about RE-Source [here](#).

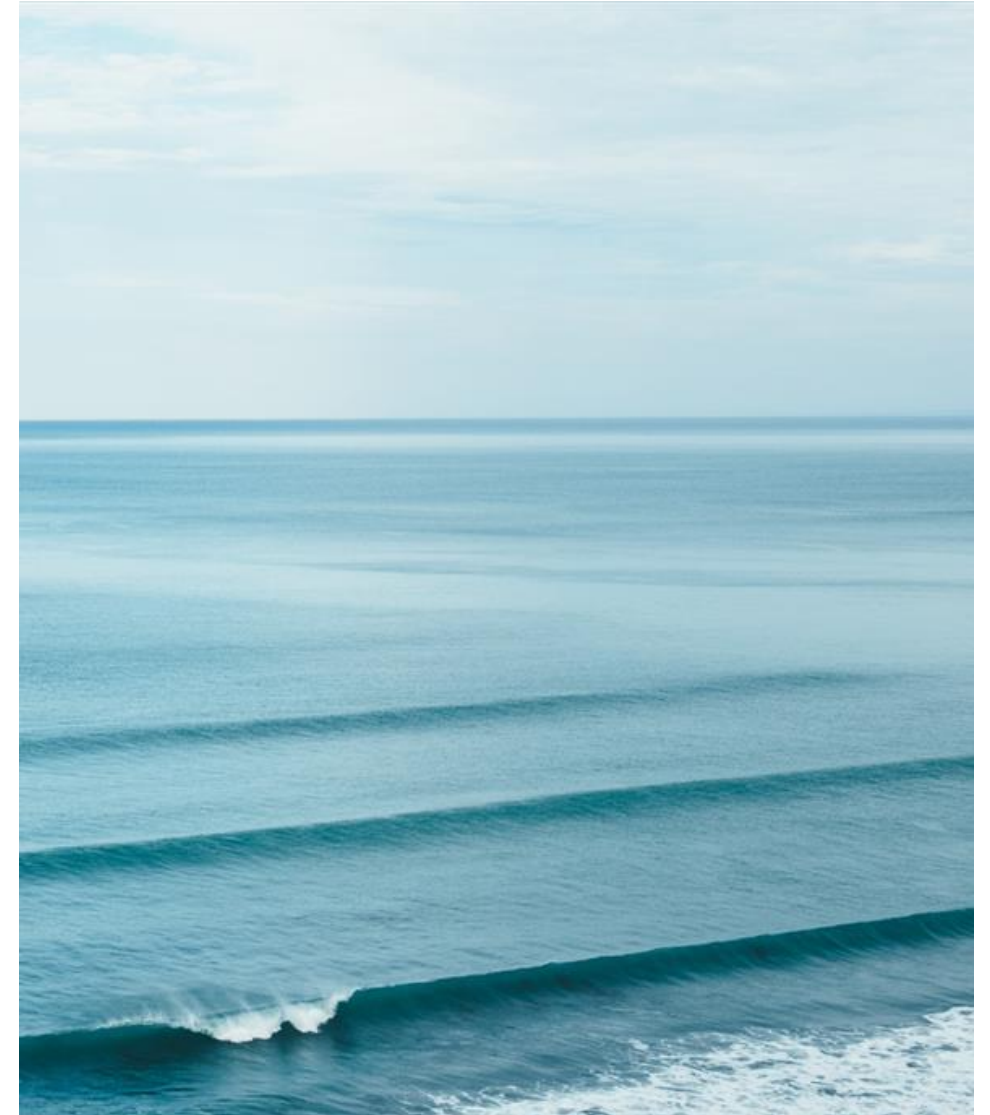
2. EFET Template

The EFET (*European Federation of Energy Traders*) has published a standard Corporate Power Purchase Agreement in cooperation with RE-Source Platform. In view of standardization, industry stakeholders have widely circulated and reviewed this CPPA.

Check out Pexapark's commentary on EFET's Sample PPA Contract [here](#)
Download EFET's Sample PPA Contract [here](#)

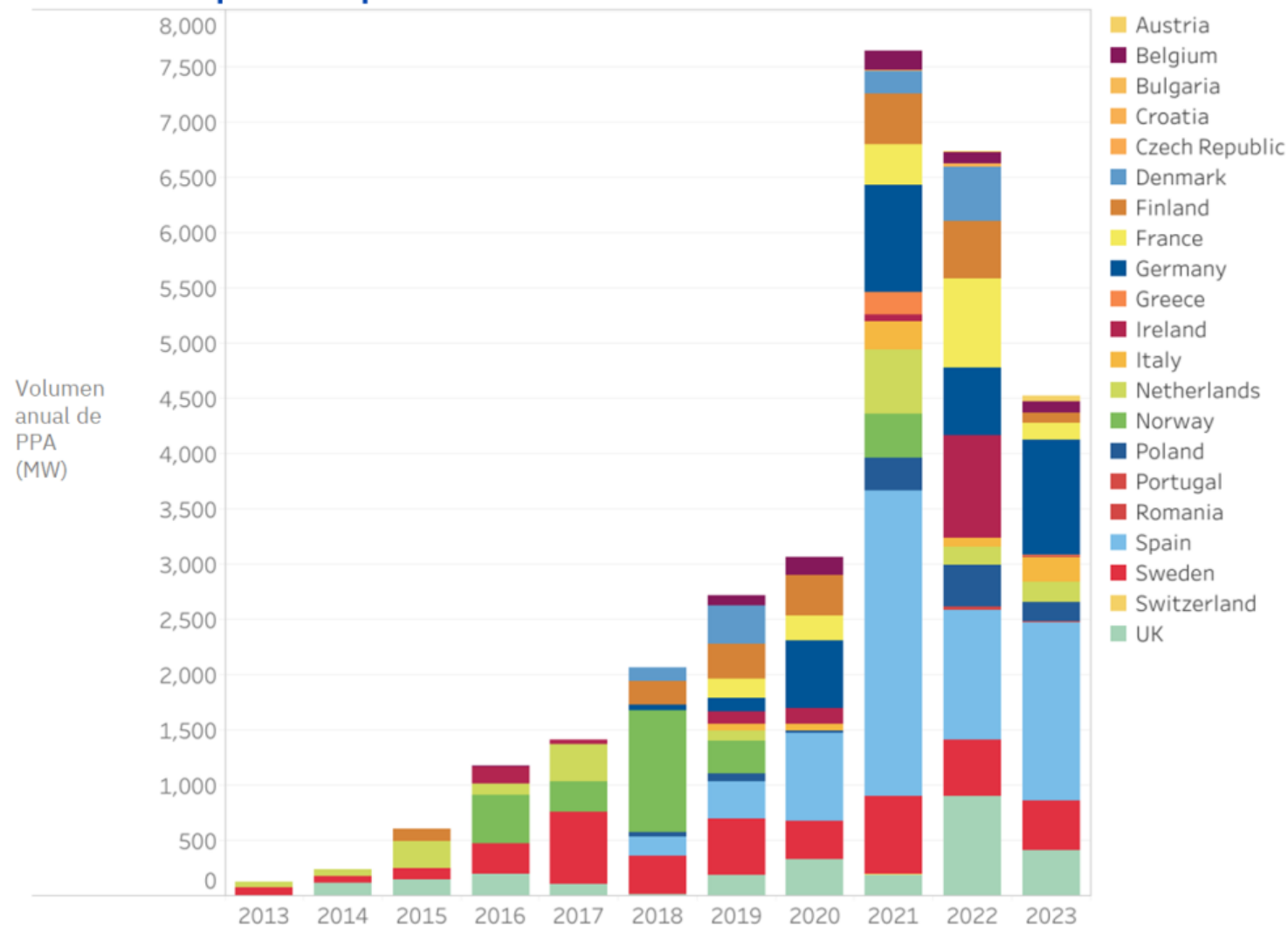
Contents

- 01 PPAs introduction
- 02 Renewable PPAs
 - > Parties involved
 - > Objectives of the parties
 - > Types of PPAs
 - > Contractual information
 - > Risk in PPAs
- 03 PPA process
- 04 Digital platforms for PPAs
- 05 PPAs trends and market outlook



PPA en Europa

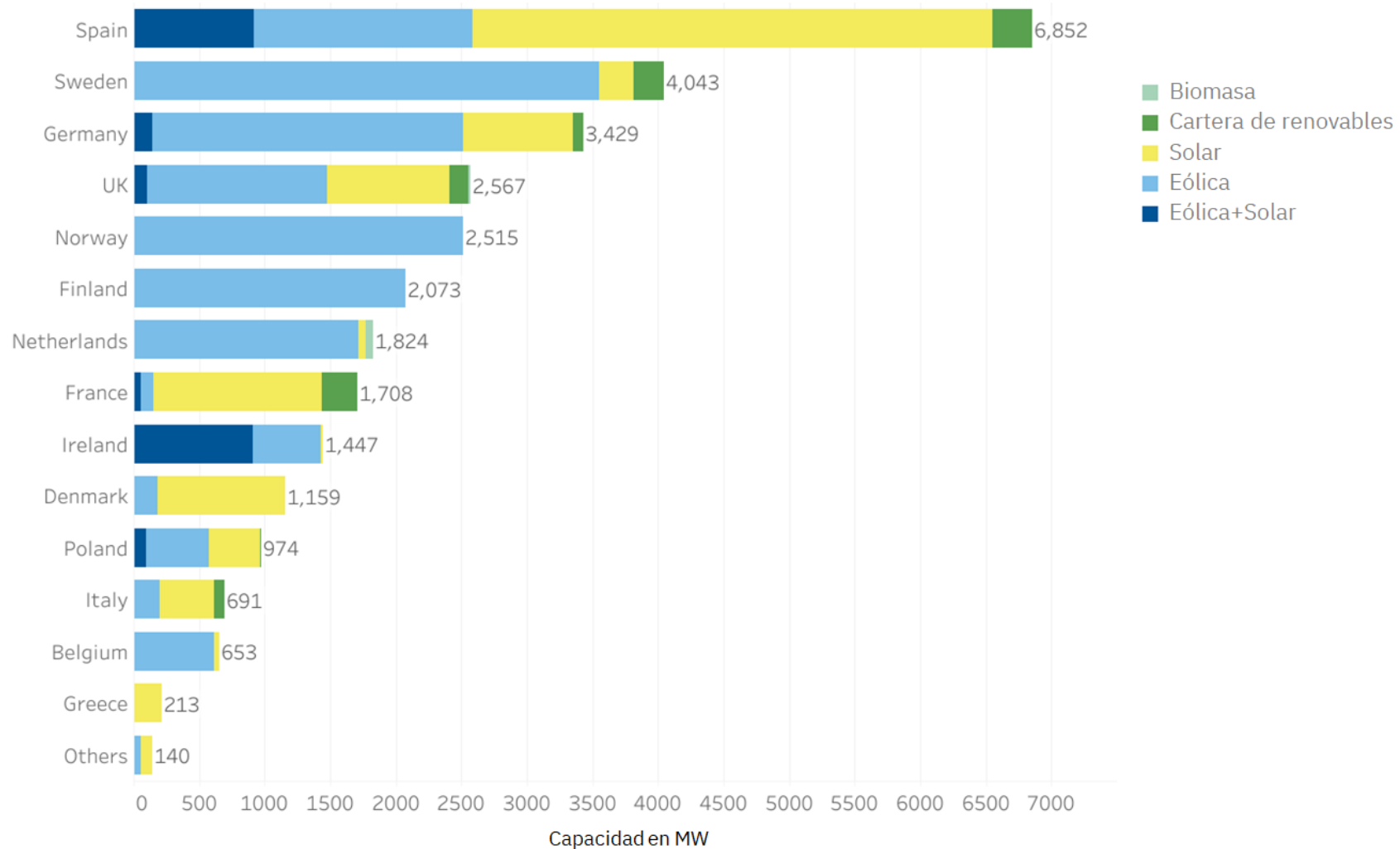
PPA por países



Fuente: WindEurope

PPA en Europa

PPA por país y tecnología



Fuente: WindEurope

PPA en Europa

PPA precios

LevelTen
Energy

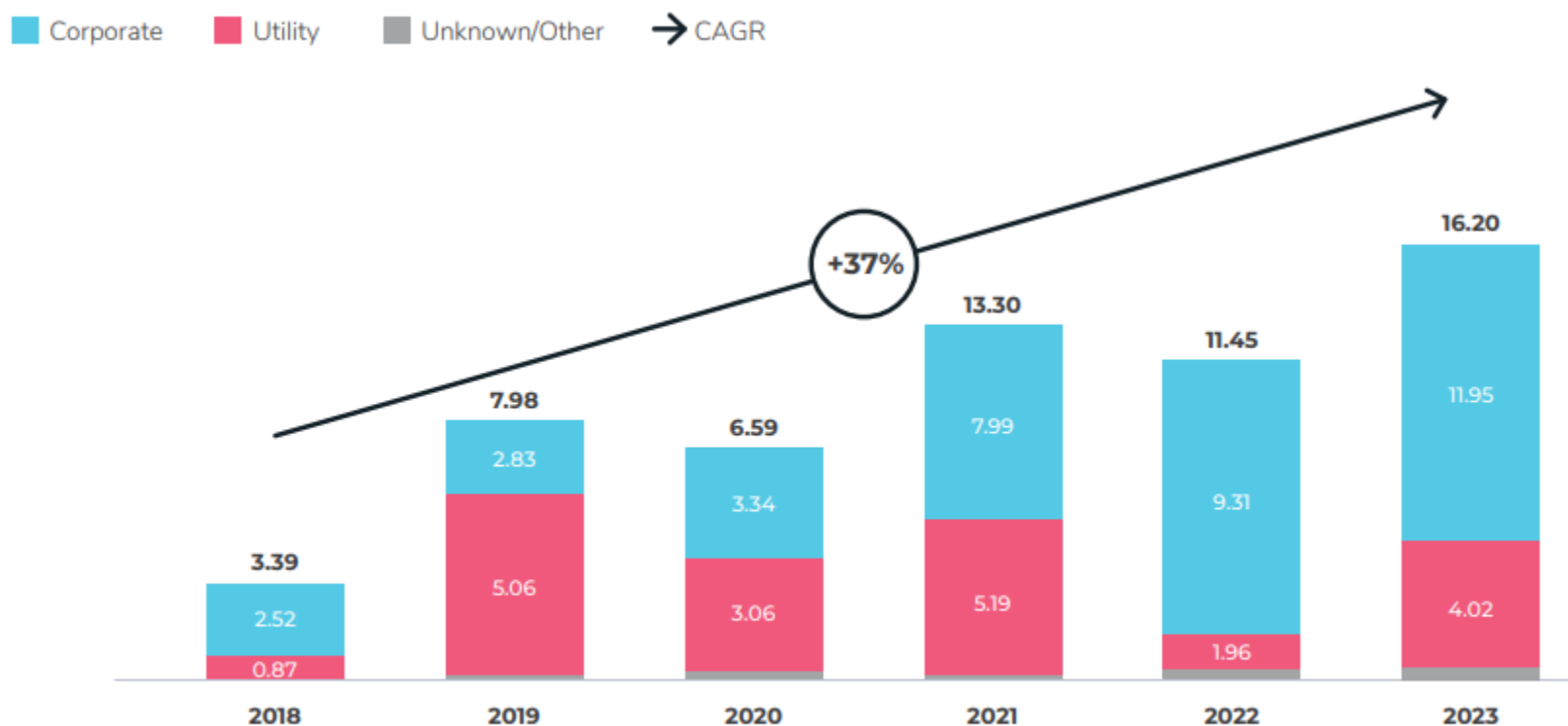
Q2 2023 Europe P25 PPA Price Index



Market- Averaged Continental Index: Q2 2021 to Q2 2023

Source: LevelTen Energy

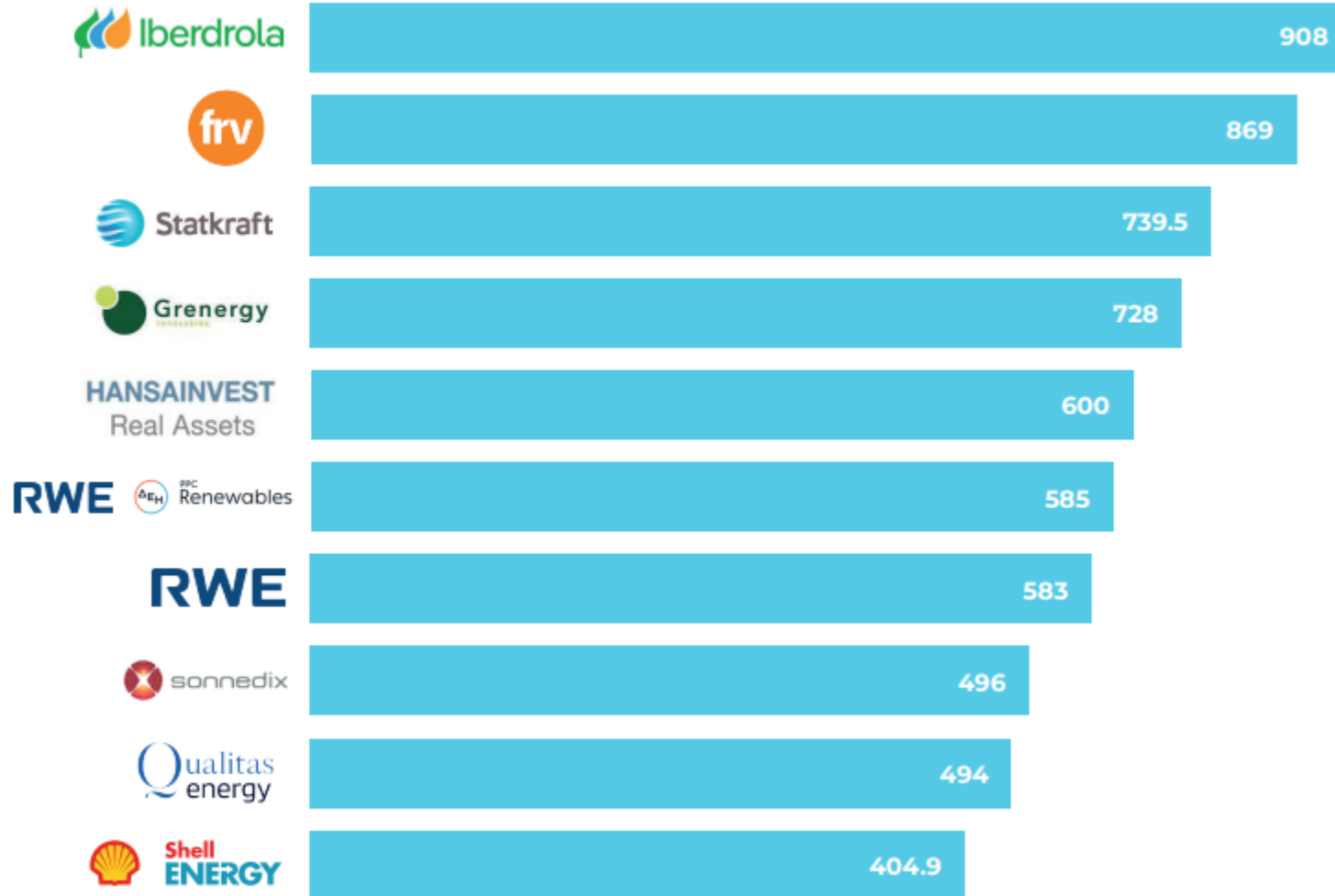
La tendencia es clara en los últimos años, ya que, **la tasa de crecimiento anual compuesta es de 37%**. En 2018 se notificaron 3,4 GW de energía, en cambio, en 2023 la cifra ascendió a 16,2 GW.



Source: [PexaQuote](#), [PPA Tracker](#) Note: 'Other' mostly refers to electrolyser developers

Flujo de acuerdos de PPA por capacidad contratada divulgada,
2018-2023 (GW)

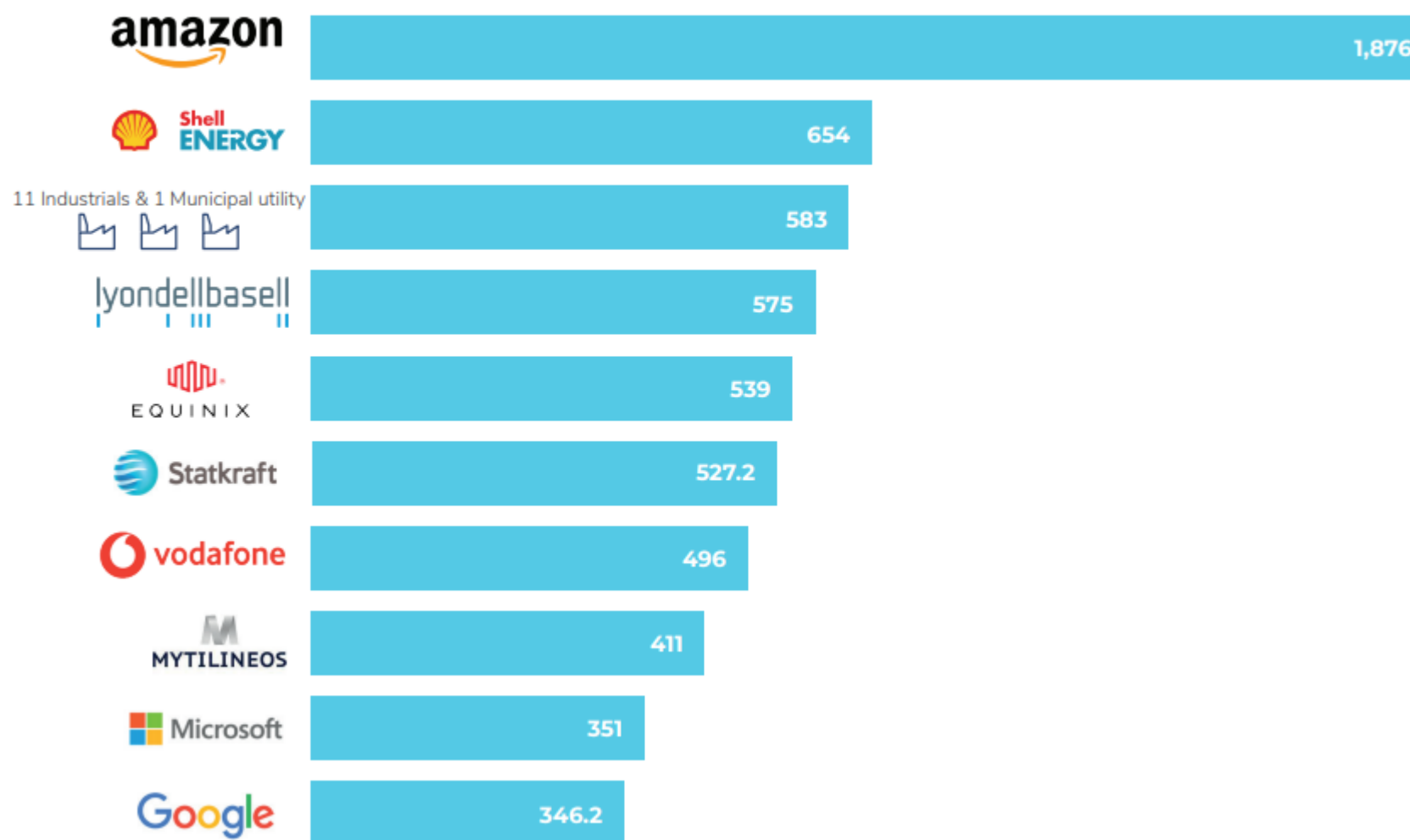
Top Sellers in 2023 by volumes (MW)



Source: [PexaQuote](#), [PPA Tracker](#)

Las empresas con más PPAs vendidos en 2023 por volúmenes (MW)

Top Buyers in 2023 by volumes (MW)



Principales compradores en 2023 por volumen (MW)

ANO	PAIS	MODALIDAD	CONTRAPARTES	SUMINISTRO	DURACIÓN
2020	España	Generador-cliente corporativo VPPA	Baywa r.e.-AB InBev	250 GWh/año	10 años
		Generador-comercializadora	Fortia-Engie	400 GWh/año	11 años
		Generador-cliente corporativo Físico Off-site	Iberdrola-Orange	200 GWh/año	12 años
		Generador-cliente corporativo Físico Off-site	Acciona-Telefónica	100 GWh/año	10 años
		Generador-cliente corporativo	Sonnedit-Statkraft	100 GWh/año	10 años
		Generador-cliente corporativo Físico On-site	Iberdrola-Makro	6 MW	15 años
		Generador-cliente corporativo VPPA	EDP Renovables-Novartis	63 MW	15 años
		Generador-cliente corporativo VPPA	Acciona-Novartis	135 MW	10 años
		Generador-cliente corporativo VPPA	Enel Green-Novartis	78 MW	10 años
		Generador-cliente corporativo	Endesa-Cellnex	300 GWh/año	10 años
		Generador-cliente corporativo VPPA	EDP Renovables-Royal DSM	59 MW	no disponible
2021	España	Generador-cliente corporativo Físico Off-site	Iberdrola-Danone	73 GWh/año	10 años
		Generador-cliente corporativo VPPA	Solaria-Shell	300 MW	10 años
	Portugal	Generador-cliente corporativo Físico Off-site	EDP Renovables-NOS	62 GWh/año	10 años
2022	España	Generador-cliente corporativo Físico	ALCOA-Greenalia	183 MW	10 años
		Generador-cliente corporativo	ALCOA-Endesa	1.151 GWh/año	10 años

Ejemplos de contratos PPAs firmados en España y Portugal.
Periodo 2020-2022

Fuente: Informe “Bird & Bird & Corporate PPAs.
An International Perspective. 2020/2021 Edition”

05 PPAs trends and market outlook



The 10 most recent PPA deals:

Announcement Date	Country	Buyer	Seller	Technology	PPA Size (MW)	Project Size (MW)	PPA Tenor (years)	Offtaker Type	Corporate Offtaker	Storage Capacity (MW)
30 Sep 2025	Spain	Turn2X	Axpo	?				Corporate	Green hydrogen producer	
30 Sep 2025	Spain	Fortia Energia	Statkraft	1 ☀			5	Corporate	Metals and Mining	
24 Sep 2025	Bulgaria	Teva Pharmaceutical Industries	Enery	☀	36.6	122	15	Corporate	Healthcare	
22 Sep 2025	Poland	KGHM Polska Miedź	Engie	1 ☀	30.75	30.75		Corporate	Metals and Mining	
22 Sep 2025	United States	Amazon	Avangrid	☀	57	57		Corporate	Information Technology	
18 Sep 2025	Germany	Mercedes-Benz	Umweltgerechte Kraftanlagen GmbH & Co. KG (UKA)	1 ☀	140	140	25	Corporate	Automobiles and Components	
16 Sep 2025	Germany	Stellantis	Athos Solar	☀	20	20		Corporate	Automobiles and Components	
16 Sep 2025	United States	Daikin	Engie	☀		260	5	Corporate	Information Technology	
15 Sep 2025	Spain	Leroy Merlin	NTR	☀	26.4	49.6	10	Corporate	Consumer Staples	
11 Sep 2025	Great Britain	Unite Students	Blackfinch Energy/Engie	☀	25	25	5	Corporate	Educational Services	

July 29, 2025

EE ESPAÑA, Almacenamiento, Eólica

 Spain

TotalEnergies propone una evolución de los PPA para impulsar el crecimiento de la electrificación y la demanda en España

En el marco de FES Iberia 2025, Gonzalo Barba, Managing Director de TotalEnergies, remarcó la necesidad de adaptar los contratos PPA al nuevo contexto energético. Además, llamó a incrementar la electrificación, atraer inversión industrial y agilizar la regulación para mantener el liderazgo renovable de España.

La bajada de precios de los PPA solares en España amenaza la rentabilidad de los proyectos, según LevelTen

En el primer trimestre de 2025, los precios solares P25 en España descendieron un 5%, aproximadamente 2 €/MWh, y los desarrolladores están alcanzando los límites inferiores de los precios de los PPA necesarios para obtener la rentabilidad requerida.

MAYO 13, 2025 **PILAR SÁNCHEZ MOLINA**

Los precios de los PPAs suben un 4,2% en España en junio, según Pexapark

La consultora suiza afirma que el mercado europeo se recuperó con fuerza en junio tras los malos resultados de mayo, con 20 PPAs por un total de 1.429 MW.

JULIO 21, 2025 **PILAR SÁNCHEZ MOLINA**

Gracias!

